Growing Rice in a UK Greenhouse

Experimental project: yield, cost and site options

Summary

This project tests whether short-season temperate japonica rice can be grown in a UK greenhouse. The recommended size is **1/100 acre** (about 40 m^2, roughly 30 m^2 usable), which holds about 750 plants and gives about 4.5 kg of brown rice in an average season. It is a worthwhile learning project, but it will not compete with imported rice at £1.50 to £3 per kg.

All figures are my own rough estimates in GBP. Very little published data exists for UK greenhouse rice.

1. Growing plan

Rice is not grown commercially in the UK because the climate is too cool and the summers too short. In a greenhouse, short-season varieties (100 to 120 days to maturity) are the best option. Buy untreated seed meant for sowing; supermarket rice will not germinate.

Period	Stage	Notes
Feb - Mar	Sow	Heated propagator, 25 to 30°C
Apr - May	Seedlings	Keep warm, add grow lights
Late May	Transplant	When 15 to 20 cm tall
Jul - Aug	Flowering	Critical stage, needs warmth
Sep - Oct	Harvest	Heads golden, grains firm

- **Containers:** tubs or lined troughs with no drainage holes, so they can stay flooded.
- **Soil:** heavy loam or garden soil with compost; avoid light peaty mixes.
- **Water:** keep 3 to 5 cm standing over the soil; warm rainwater is best.
- **Climate:** days 25 to 32°C, nights above 15 to 18°C; shade above about 35°C during flowering.
- **Spacing:** about 20 x 20 cm (25 plants per m^2).

2. Experiment design

Use 4 varieties with 3 replicate plots each, every plot about 2.5 m^2 (about 62 plants), totalling about 30 m^2. Change one variable at a time and randomise plot positions. Record sowing, transplant and flowering dates, daily minimum and maximum temperature, grain filling and yield per plot.

3. Expected yield

Season (30 m^2 usable)	Paddy rice	Brown rice (~75%)	White rice (~67%)
Poor (100 g/m^2)	3 kg	2.3 kg	2 kg
Average (200 g/m^2)	6 kg	4.5 kg	4 kg
Good (300 g/m^2)	9 kg	6.8 kg	6 kg

4. Seasonal running cost

Item	Cost
Seed	£10 - £30
Soil and compost (~4.5 m^3, mostly first year)	£200 - £400
Heating (spring and autumn)	£300 - £800
Water (10 to 20 m^3)	£30 - £60
Fertiliser	£30 - £60
Pest control and miscellaneous	£30 - £60
Total	£600 - £1,400

Assumes a greenhouse is already owned. A new polytunnel of about 40 m^2 would add roughly £1,500 to £4,000, and beds and propagation equipment £400 to £850. Labour of 100 to 150 hours is not included.

5. Does more land reduce the cost?

Only up to a point. Labour falls sharply with scale, but heating, inputs and structure grow with every square metre covered. Cost per kg levels off at about £95, roughly 38 times the import price of about £2.50 per kg.

Land	Brown rice per year	Cost per kg	Multiple of import price
1/100 acre	~5 kg	~£221	~88x
1/10 acre	~46 kg	~£164	~65x
1 acre	~455 kg	~£107	~43x
10 acres	~4.6 tonnes	~£96	~38x
100 acres	~46 tonnes	~£95	~38x
1,000 acres	~455 tonnes	~£95	~38x

To match imports at £2.50 per kg, a heated greenhouse would need yields about 32 times higher or costs about 32 times lower; an unheated polytunnel is still about 8 times away. Realistic routes are premium pricing (£60 to £150 per kg for specialty UK-grown rice) or treating the project as a hobby or educational activity.

6. Where to look for growing space

These are places to ask, not confirmed rentals. Ask about rent for about 40 m^2 from March to October, water and power costs, and whether flooded beds are allowed.

Place	Area	Phone
Lea Valley Growers Association	Cheshunt, Waltham Cross	+44 1992 625076
Cottage Hedging	Stapleford Tawney, near Romford	+44 7500 110471
Connaught Farm Nurseries	Loughton	+44 20 8504 5168
Grenville Nurseries	Writtle, Chelmsford	+44 1245 420400
ARU Writtle (horticulture college)	Chelmsford	+44 1245 424200
AE Harnett and Sons	Ingatestone	+44 1277 840726

Also worth contacting for advice: UKCEH, which ran the UK's first rice paddy trial on rewetted peat near Ely in 2025, where a Colombian variety performed best among nine tested.

7. Next steps

- Contact growers or allotment societies about 40 m^2 of covered space.
- Source untreated short-season japonica seed and check plant import rules.
- Set up 4 varieties x 3 replicates and sow in late February to March.
- Record dates, temperature and yield, then decide whether to scale to 1/10 acre.